DIVYAPRAKASH

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EDUCATION

Year	Degree	Institute	GPA / Marks(%)
2026 (Expected)	Ph.D.	Indian Institute of Technology Delhi	9
2016	M.Tech.	Indian Institute of Technology Gandhinagar	8.11
2012	B.E.	Dr. Ambedkar Institute of Technology, Bangalore	76

TECHNICAL SKILLS

- **Programming: C/C++**, Python, Fortran, MATLAB (proficient); CUDA, Git, Bash, Linux (experienced)
- **Numerical Methods & HPC:** FVM, FEM, finite-difference and projection methods, Immersed Boundary Method; coupled **fluid–structure** solver development; **multiphase** CFD (VOF); linear equation solvers; **GPU acceleration** (NVIDIA AmgX)
- **Modeling & Simulation:** OpenFOAM, ANSYS Fluent, Star-CCM+, ParaView, gmsh (proficient/experienced); SU2, FDS (experienced)
- **AI/ML:** PyTorch, TensorFlow (experienced); **machine learning for scientific computing** (applications to CFD)

WORK EXPERIENCE

- **Convergent Science, Pune, India**, Research Engineer, Development, **(Oct, 2025 – Present)**: Core development of **fluid–structure interaction** in **CONVERGE**, a commercial CFD code, in **object-oriented C++**; implemented new physical models, delivered **solver speedups**, and improved the robustness of existing numerical models.
- **Jorhat Engineering College, Assam, India**, Assistant Professor, **(Sept, 2018 – Sept, 2020)**
- **Institute of Technology Gopeshwar, Uttarakhand, India**, Assistant Professor, **(Jan, 2018 – Aug, 2018)**
- **Indian Institute of Technology Gandhinagar, Gujarat, India**, Junior Research Fellow, **(Sept, 2016 – Aug, 2017)**
- **Institute for Plasma Research Gandhinagar, Gujarat, India**, Graduate Apprentice, **(Aug, 2013 – Nov, 2013)**

SELECTED PROJECTS

- **GPU-Accelerated FSI Solver (2024)** : Developed a fluid–structure interaction solver using the Immersed Boundary Method (IBM) on a staggered finite-difference grid, with the **pressure Poisson equation GPU-accelerated** via **NVIDIA AmgX**. Implemented in Fortran and C, with MATLAB and Python for postprocessing and visualization. Code
- **Towards High-Fidelity Modeling of Superdroplet Condensation using DNS Data and Machine Learning (2023 – 2025)**: Developed a machine learning-based parameterization for droplet growth in LES using high-fidelity DNS data. Paper Code
- **Vortices in a Stirred Coffee Cup (2014)**: Performed 3D **multiphase** CFD (VOF + DFBI) in Star-CCM+ with overset mesh to simulate vortex formation from a stirrer and spoon, analyzing vorticity, wall shear stress, and free-surface dynamics.

Full Project Portfolio

INTERNSHIPS

- **Google Summer of Code 2025, Remote (May - Sept, 2025)** : Stitching SU2 – Using data-driven, physics-informed machine learning to model fluid properties in CFD. **Report** *(Selected among $\sim 8\%$ of applicants worldwide)*
- **Institute of High Performance Computing, Singapore (May – July, 2015)** : Implementation of IBM in OpenFOAM

IIT DELHI PHD THESIS

Title: Sensing and Transport of Particles Using Arrays of Artificial Cilia
Supervisor: Amitabh Bhattacharya
Description: Cilia, hair-like organelles for fluid transport, locomotion, and sensing, are modeled via Kirchhoff rods and particles via FEM, in a 2D immersed boundary framework of cilia–fluid–deformable particle interactions and extended to a 3D **OpenFOAM** coupled setup. A novel, non-optical method using an **LSTM** neural network infers particle size and shape from cilia deflections, and simulations show how ciliary peristalsis drives particle transport.

SELECTED PUBLICATIONS

- **A Review of Computational Modeling of Fluid-Immersed Flexible Filaments**, *Journal of the Indian Institute of Science*, 104:277, 2024. DOI: 10.1007/s41745-024-00423-x
- **Machine Learning Based Sensing of Particle Shape and Size Using Passive Artificial Cilia**, *Journal of Fluids and Structures*, 140:104446, 2026. DOI: 10.1016/j.jfluidstructs.2025.104446

Google Scholar (ID: H4Byf9EAAAAJ)

CONFERENCES

- **preCICE Workshop 2024, University of Stuttgart, Germany (Sept 2024)**: Poster Presentation – Awarded Travel Grant
- **ECCOMAS Congress 2024, Lisbon, Portugal (June 2024)**: Oral Presentation – Art & Science Competition Winner
- **Workshop on Interfacial Engineering at Multiple Spatio-temporal Scales, IISc Bengaluru (Jan 2024)**: Best Poster Award

LEADERSHIP & ACTIVITIES

- **Program Coordinator (Volunteer), Indiahikes, (Oct 2023):** Coordinated an experiential Himalayan trek for IIM Indore students, fostering leadership skills amidst challenging terrain and weather at 12,250 ft (Brahmatal, Uttarakhand)